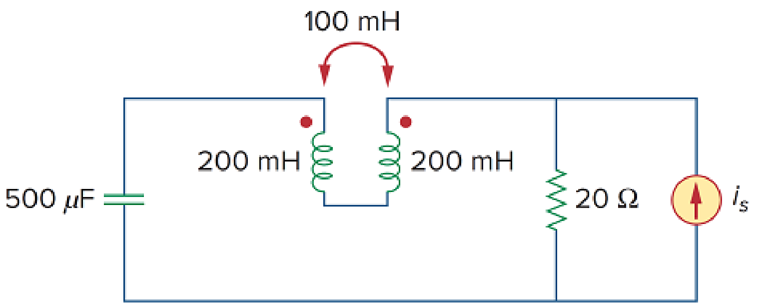
Chapter 13, Problem 23P

Problem

Let *is* = 5 cos (100*t*) A. Calculate the voltage across the capacitor, *vc*. Also calculate the value of the energy stored in the coupled coils at *t* = 2.5*π* ms.

Figure 13.92



Step-by-step solution

Step 1 of 8

Refer to Figure 13.92 in the textbook.

Determine the nature of connection of the two coils using dot convention.

Since the current is entering the dot of the first coil, the polarity of the mutual voltage is positive at the dot of the second coil.

Since the current is leaving the dot of the second coil, the polarity of the mutual voltage is negative at the dot of the first coil.

Thus, the two coils are in series opposition as shown in Figure 1.

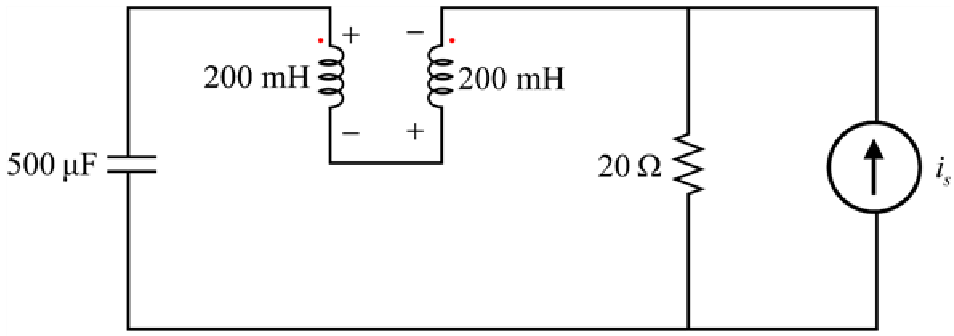


Figure 1

Step 2 of 8

Write the expression for the equivalent inductance, L_{eq} for two coils in series opposition.

$$L_{eq} = L_1 + L_2 - 2M$$

Substitute 200 mH for L_1, 200 mH for L_2 and 100 mH for M.

$$\begin{aligned} L_{eq} &= 200 + 200 - 2(100) \\ &= 200\text{ mH} \end{aligned}$$

Step 3 of 8

Use the Current to Voltage Source Transformation to transform the current source into a voltage source (V_s) with $20\ \Omega$ resistor in series with it.

$$V_s = i_s R$$

Substitute $i_s = 5\cos(100t)$ and $20\ \Omega$ for R.

$$V_s = [5\cos(100t)](20)$$

$$= 100\cos(100t)\ \text{V} \quad \ldots\ldots (1)$$

$$= 100\angle 0°\ \text{V}$$

Write the standard equation for a voltage source.

$$V_s = V_m \cos\omega t \quad \ldots\ldots (2)$$

Compare equation (1) and equation (2) to obtain the value of ω.

$$\omega = 100\ \text{rad/s}$$

Step 4 of 8

Calculate the inductive impedance of the series equivalent of the two coils, $X_{L_{eq}}$.

$Z_{L_{eq}} = j\omega L_{eq}$ Substitute $200\ \text{mH}$ for L_{eq} and $100\ \text{rad/s}$ for ω.

$$Z_{L_{eq}} = j(100)(200\times10^{-3})$$
$$= j20\ \Omega$$

Calculate the capacitive impedance, X_C.

$Z_C = \dfrac{1}{j\omega C}$ Substitute $500\ \mu\text{F}$ C and $100\ \text{rad/s}$ for ω.

$$Z_C = \frac{1}{j(100)(500\times10^{-6})}$$
$$= \frac{1}{j5\times10^{-2}}$$
$$= -j20\ \Omega$$

Step 5 of 8

Sketch the modified circuit as shown in Figure 2.

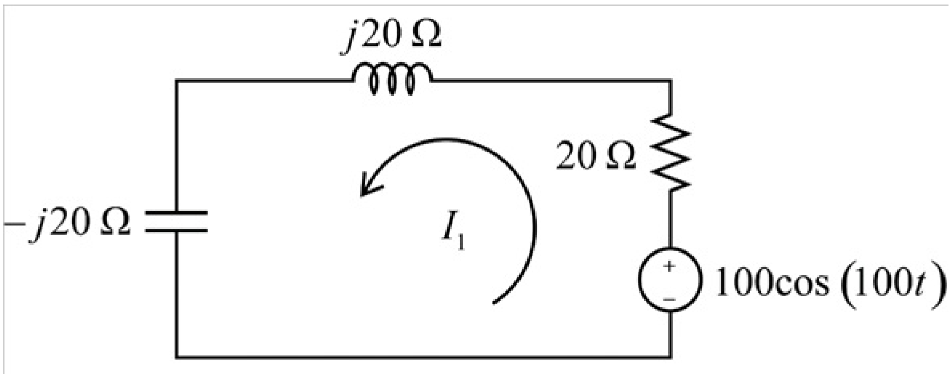


Figure 2

Step 6 of 8

Apply Kirchhoff's voltage Law in the loop.

$$100\angle 0° = 20I_1 + j20I_1 - j20I_1$$
$$I_1 = \frac{100\angle 0°}{20}$$
$$I_1 = 5\angle 0° \text{ A}$$
$$I_1 = 5\cos(100t) \text{ A}$$

Calculate the voltage across the capacitor, v_C.

$$v_C = Z_C I_1$$

Substitute $5\angle 0° \text{ A}$ for I_1 and $-j20\,\Omega$ for Z_C.

$$\begin{aligned} v_C &= (-j20)(5\angle 0°) \\ &= -j100 \\ &= 100\angle -90° \text{ V} \end{aligned}$$

Represent v_C in sinusoidal form.

$$v_C = 100\cos(100t - 90°) \text{ V}$$

Hence, the voltage across the capacitor, v_C is $\boxed{100\cos(100t - 90°) \text{ V}}$.

Step 7 of 8

Calculate the instantaneous value (i_1) of the current I_1 at time $t = 2.5\pi \text{ ms}$.

$$\begin{aligned} i_1 &= 5\cos(100t) \\ &= 5\cos(100 \times 2.5 \times \pi \times 10^{-3}) \\ &= 5 \text{ A} \end{aligned}$$

Write the expression for the energy stored in a coupled circuit, w.

$$w = \frac{1}{2}L_1 i_1^2 + \frac{1}{2}L_2 i_2^2 - M i_1 i_2$$

Since same current passes through both the coils, therefore,

$$w = \frac{1}{2}L_1 i_1^2 + \frac{1}{2}L_2 i_1^2 - M i_1^2$$

Step 8 of 8

Substitute 200 mH for L_1 and L_2, 100 mH for M and 5 A for i_1.

$$\begin{aligned} w &= \frac{1}{2}(200 \times 10^{-3})(5)^2 + \frac{1}{2}(200 \times 10^{-3})(5)^2 - (100 \times 10^{-3})(5)^2 \\ &= 2.5 + 2.5 - 2.5 \\ &= 2.5 \text{ J} \end{aligned}$$

Hence, the energy stored in the coupled coils, w is $\boxed{2.5 \text{ J}}$.